

lab05(SSU)

September 17, 2025

1 Lab 5: Table Manipulation and Visualization

Helpful Resource: - [Python Reference](#): Cheat sheet of helpful array & table methods!

Recommended Reading: * [Visualization](#)

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
      !pip install py pandoc
      !pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

The following additional packages will be installed:

```
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronet2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzzip-0-13 lmodern pandoc-data poppler-data preview-latex-style
rake ruby ruby-net-telnet ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0
rubygems-integration tlutils teckit tex-common tex-gyre texlive-base
texlive-binaries texlive-fonts-recommended texlive-latex-base
texlive-latex-recommended texlive-pictures texlive-plain-generic tipa
xfonts-encodings xfonts-utils
```

Suggested packages:

```
fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
texlive-luatex pandoc-citeproc context wkhtmltopdf librsvg2-bin groff ghc
nodejs php python libjs-mathjax libjs-katex citation-style-language-styles
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-pstricks
```

```

dot2tex prerex texlive-pictures-doc vprerex default-jre-headless tipa-doc
The following NEW packages will be installed:
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm.3 libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronet2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzzip-0-13 lmodern pandoc pandoc-data poppler-data
preview-latex-style rake ruby ruby-net-telnet ruby-rubygems ruby-webrick
ruby-xmlrpc ruby3.0 rubygems-integration t1utils teckit tex-common tex-gyre
texlive texlive-base texlive-binaries texlive-fonts-recommended
texlive-latex-base texlive-latex-extra texlive-latex-recommended
texlive-pictures texlive-plain-generic texlive-xetex tipa xfonts-encodings
xfonts-utils
0 upgraded, 58 newly installed, 0 to remove and 35 not upgraded.
Need to get 202 MB of archives.
After this operation, 728 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security/main amd64 ruby-rubygems
all 3.3.5-2ubuntu1.2 [228 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.12 [753 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.12 [5,031 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64
2.13.1-1 [1,221 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-lmodern all
2.004.5-6.1 [4,532 kB]

```

Get:16 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all 20201225-1build1 [397 kB]
Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]
Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java all 18-1 [4,720 B]
Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-gfm0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [115 kB]
Get:20 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-gfm-extensions0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [25.1 kB]
Get:21 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-java all 43-1 [10.8 kB]
Get:22 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]
Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]
Get:25 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.11 [50.1 kB]
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1 [5,100 B]
Get:27 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7 kB]
Get:28 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all 0.1.1-2 [12.6 kB]
Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick all 1.7.0-3ubuntu0.2 [52.5 kB]
Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all 0.3.2-1ubuntu0.1 [24.9 kB]
Get:31 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
Get:33 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64 2.5.11+ds1-1 [421 kB]
Get:34 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
Get:35 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
Get:36 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64 0.13.72+dfsg.1-1.1 [27.0 kB]
Get:37 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all 1:1.0.5-0ubuntu2 [578 kB]
Get:38 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64 1:7.7+6build2 [94.6 kB]
Get:39 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all 2.004.5-6.1 [9,471 kB]

```

Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 pandoc-data all
2.9.2.1-3ubuntu2 [81.8 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 pandoc amd64
2.9.2.1-3ubuntu2 [20.3 MB]
Get:42 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:55 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:56 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:57 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:58 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 202 MB in 4s (57.4 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126435 files and directories currently installed.)
Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...

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Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.12_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.12_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcmark-gfm0.29.0.gfm.3:amd64.

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Preparing to unpack .../17-libcmark-gfm0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...
Unpacking libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcmark-gfm-
extensions0.29.0.gfm.3:amd64.
Preparing to unpack .../18-libcmark-gfm-
extensions0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...
Unpacking libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../19-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../20-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../21-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../22-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../23-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../24-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../25-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../26-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../27-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../28-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../29-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../30-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntaxtex2:amd64.
Preparing to unpack .../31-libsyntaxtex2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntaxtex2:amd64 (2021.20210626.59705-1ubuntu0.2) ...

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Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../32-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../33-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../34-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../35-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../36-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../37-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../38-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package pandoc-data.
Preparing to unpack .../39-pandoc-data_2.9.2.1-3ubuntu2_all.deb ...
Unpacking pandoc-data (2.9.2.1-3ubuntu2) ...
Selecting previously unselected package pandoc.
Preparing to unpack .../40-pandoc_2.9.2.1-3ubuntu2_amd64.deb ...
Unpacking pandoc (2.9.2.1-3ubuntu2) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../41-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../42-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../43-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../44-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../45-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../46-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...

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Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../47-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../48-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../49-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../50-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../51-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../52-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../53-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../54-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../55-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../56-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../57-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...

```

```

Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up pandoc-data (2.9.2.1-3ubuntu2) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up pandoc (2.9.2.1-3ubuntu2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex

```

```
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link
```

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

Processing triggers for tex-common (6.17) ...

Running upmap-sys. This may take some time... done.

Running mktexlsr /var/lib/texmf ... done.

Building format(s) --all.

This may take some time... done.

Collecting py pandoc

Downloading py pandoc-1.15-py3-none-any.whl.metadata (16 kB)

Downloading py pandoc-1.15-py3-none-any.whl (21 kB)

Installing collected packages: py pandoc

Successfully installed py pandoc-1.15

Collecting data science

Downloading data science-0.18.0-py3-none-any.whl.metadata (910 bytes)

Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from data science) (0.20.0)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from data science) (75.2.0)

Requirement already satisfied: matplotlib>=3.0.0 in /usr/local/lib/python3.12/dist-packages (from data science) (3.10.0)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from data science) (2.2.2)

Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from data science) (1.16.1)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from data science) (2.0.2)

Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-packages (from data science) (7.34.0)

Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (from data science) (5.24.1)

Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages (from data science) (0.8.1)

Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->data science) (3.1.6)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->data science) (2.32.4)

Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-packages (from folium>=0.9.1->data science) (2025.4.0)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->data science) (1.3.3)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->data science) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->data science) (4.59.2)

Requirement already satisfied: kiwisolver>=1.3.1 in

```

/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(1.4.9)
Requirement already satisfied: packaging>=20.0 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(25.0)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
packages (from matplotlib>=3.0.0->datascience) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(3.2.3)
Requirement already satisfied: python-dateutil>=2.7 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(2.9.0.post0)
Collecting jedi>=0.16 (from ipython->datascience)
  Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)
Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.4.2)
Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (0.7.5)
Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (5.7.1)
Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in
/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)
Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (2.19.2)
Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (0.2.0)
Requirement already satisfied: matplotlib-inline in
/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.1.7)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
(0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from
jinja2>=2.9->folium>=0.9.1->datascience) (3.0.2)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-

```

```

toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.13)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.8.3)
Downloading datascience-0.18.0-py3-none-any.whl (725 kB)
725.0/725.0 kB
37.6 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
32.6 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.0 jedi-0.19.2

```

```

[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')

```

Mounted at /content/drive

```

[17]: # Don't change this cell; just run it.
import numpy as np
from datascience import *
import warnings
warnings.simplefilter('ignore', FutureWarning)

# These lines do some fancy plotting magic.\n",
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plt
plt.style.use('fivethirtyeight')

```

1.1 1. Unemployment

The Great Recession of 2008-2009 was a period of economic decline observed globally, with scale and timing varying from country to country. In the United States, it resulted in a rapid rise in

unemployment that affected industries and population groups to different extents.

The Federal Reserve Bank of St. Louis publishes data about jobs in the US. Below, we've loaded data on unemployment in the United States. There are many ways of defining unemployment, and our dataset includes two notions of the unemployment rate:

1. *Non-Employment Index (or NEI)*: Among people who are able to work and are looking for a full-time job, the percentage who can't find a job.
2. *NEI-PTER*: Among people who are able to work and are looking for a full-time job, the percentage who can't find any job *or* are only working at a part-time job. The latter group is called "Part-Time for Economic Reasons", so the acronym for this index is NEI-PTER. (Economists are great at marketing.)

The source of the data is [here](#).

Question 1. The data are in a CSV file called `unemployment.csv`. Load that file into a table called `unemployment`.

Hint: After loading in the CSV file, the `unemployment` table should look like this:

```
[19]: unemployment = Table.read_table('./DS/unemployment.csv')
      unemployment
```

```
[19]: Date          | NEI      | NEI-PTER
      1994-01-01 | 10.0974 | 11.172
      1994-04-01 | 9.6239  | 10.7883
      1994-07-01 | 9.3276  | 10.4831
      1994-10-01 | 9.1071  | 10.2361
      1995-01-01 | 8.9693  | 10.1832
      1995-04-01 | 9.0314  | 10.1071
      1995-07-01 | 8.9802  | 10.1084
      1995-10-01 | 8.9932  | 10.1046
      1996-01-01 | 9.0002  | 10.0531
      1996-04-01 | 8.9038  | 9.9782
      ... (80 rows omitted)
```

Question 2. Sort the data in descending order by NEI, naming the sorted table `by_nei`. Create another table called `by_nei_pter` that's sorted in descending order by NEI-PTER instead.

```
[20]: by_nei = unemployment.sort('NEI', descending=True)
      by_nei_pter = unemployment.sort('NEI-PTER', descending=True)
```

```
[21]: # Run this cell to check your by_nei table. You do not need to change the code.
      by_nei.show(5)
```

<IPython.core.display.HTML object>

```
[22]: # Run this cell to check your by_nei_pter table. You do not need to change the
      ↪code.
      by_nei_pter.show(5)
```

<IPython.core.display.HTML object>

Question 3. Using `take`, assign `greatest_nei` to a table containing the data for the 11 quarters when NEI was greatest.

`greatest_nei` should be sorted in descending order of NEI. Note that each row of `unemployment` represents a quarter.

```
[23]: greatest_nei = by_nei.take(np.arange(11))
      greatest_nei
```

```
[23]: Date          | NEI      | NEI-PTER
      2009-10-01 | 10.9698 | 12.8557
      2010-01-01 | 10.9054 | 12.7311
      2009-07-01 | 10.8089 | 12.7404
      2009-04-01 | 10.7082 | 12.5497
      2010-04-01 | 10.6597 | 12.5664
      2010-10-01 | 10.5856 | 12.4329
      2010-07-01 | 10.5521 | 12.3897
      2011-01-01 | 10.5024 | 12.3017
      2011-07-01 | 10.4856 | 12.2507
      2011-04-01 | 10.4409 | 12.247
      ... (1 rows omitted)
```

Question 4. It's believed that many people became PTER (recall: "Part-Time for Economic Reasons") in the "Great Recession" of 2008-2009. NEI-PTER is the percentage of people who are unemployed (included in the NEI) plus the percentage of people who are PTER.

Compute an array containing the percentage of people who were PTER in each quarter. (The first element of the array should correspond to the first row of `unemployment`, and so on.)

Note: Use the original `unemployment` table for this.

```
[24]: pter = unemployment.column('NEI-PTER') - unemployment.column('NEI')
      pter
```

```
[24]: array([ 1.0746,  1.1644,  1.1555,  1.129 ,  1.2139,  1.0757,  1.1282,
             1.1114,  1.0529,  1.0744,  1.1004,  1.0747,  1.0705,  1.0455,
             1.008 ,  0.9734,  0.9753,  0.8931,  0.9451,  0.8367,  0.8208,
             0.8105,  0.8248,  0.7578,  0.7251,  0.7445,  0.7543,  0.7423,
             0.7399,  0.7687,  0.8418,  0.9923,  0.9181,  0.9629,  0.9703,
             0.9575,  1.0333,  1.0781,  1.0675,  1.0354,  1.0601,  1.01 ,
             1.0042,  1.0368,  0.9704,  0.923 ,  0.9759,  0.93 ,  0.889 ,
             0.821 ,  0.9409,  0.955 ,  0.898 ,  0.8948,  0.9523,  0.9579,
             1.0149,  1.0762,  1.2873,  1.4335,  1.7446,  1.8415,  1.9315,
             1.8859,  1.8257,  1.9067,  1.8376,  1.8473,  1.7993,  1.8061,
             1.7651,  1.7927,  1.7286,  1.6387,  1.6808,  1.6805,  1.6629,
             1.6253,  1.6477,  1.6298,  1.4796,  1.5131,  1.4866,  1.4345,
             1.3675,  1.3097,  1.2319,  1.1735,  1.1844,  1.1746])
```

Question 5. Add `pter` as a column to `unemployment` (name the column `PTER`) and sort the resulting table by that column in descending order. Call the resulting table `by_pter`.

Try to do this with a single line of code, if you can.

```
[25]: by_pter = unemployment.with_columns('PTER', pter).sort('PTER', descending=True)
      by_pter
```

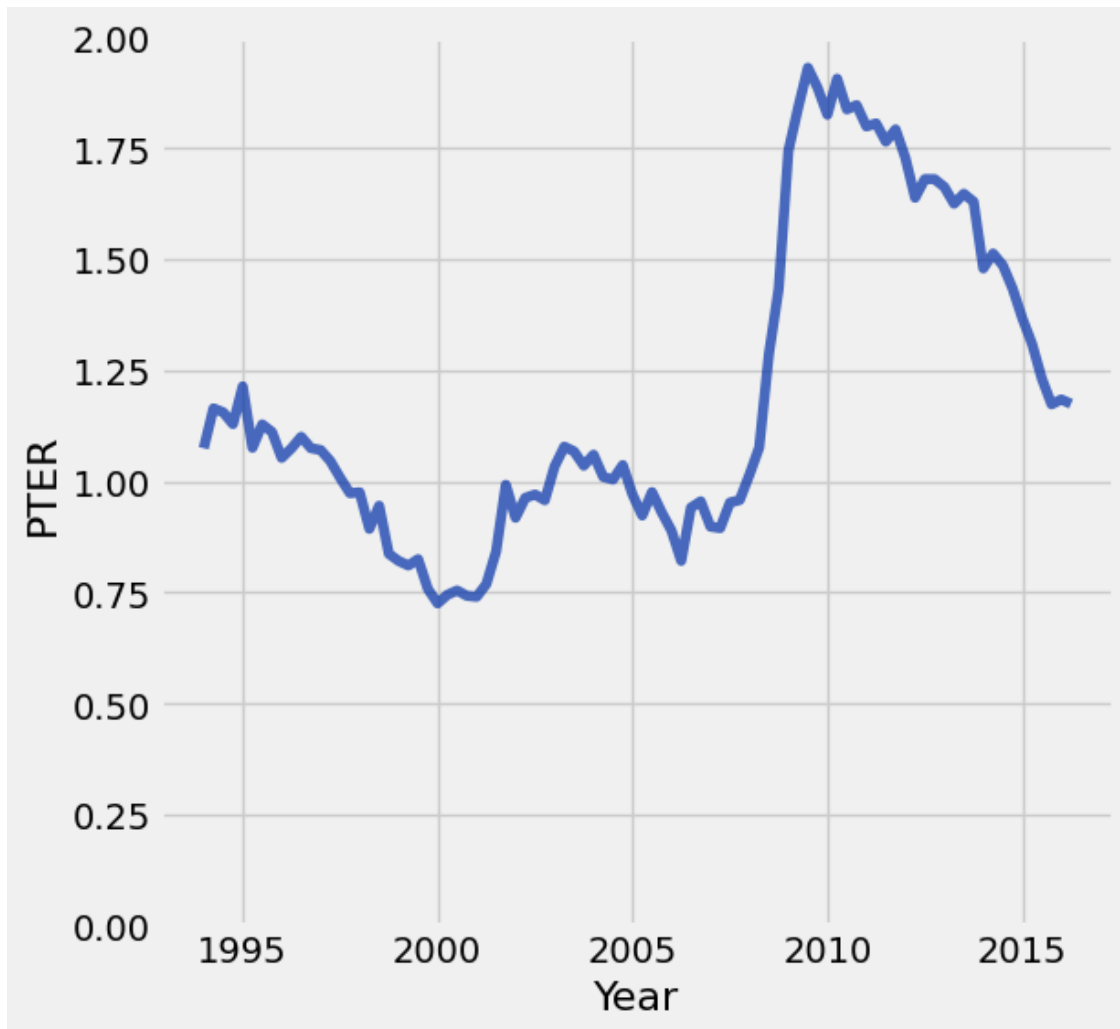
```
[25]: Date          | NEI      | NEI-PTER | PTER
      2009-07-01 | 10.8089 | 12.7404 | 1.9315
      2010-04-01 | 10.6597 | 12.5664 | 1.9067
      2009-10-01 | 10.9698 | 12.8557 | 1.8859
      2010-10-01 | 10.5856 | 12.4329 | 1.8473
      2009-04-01 | 10.7082 | 12.5497 | 1.8415
      2010-07-01 | 10.5521 | 12.3897 | 1.8376
      2010-01-01 | 10.9054 | 12.7311 | 1.8257
      2011-04-01 | 10.4409 | 12.247  | 1.8061
      2011-01-01 | 10.5024 | 12.3017 | 1.7993
      2011-10-01 | 10.3287 | 12.1214 | 1.7927
      ... (80 rows omitted)
```

Question 6. Create a line plot of `PTER` over time. To do this, create a new table called `pter_over_time` with the same columns as the `unemployment` table with the addition of two new columns: `Year` and `PTER` using the `year` array and the `pter` array, respectively. Then, generate a line plot using one of the table methods you've learned in class.

The order of the columns matter for our correctness tests, so be sure `Year` comes before `PTER`.

Note: When constructing `pter_over_time`, do not just add the `year` column to the `by_pter` table. Please follow the directions in the question above.

```
[26]: year = 1994 + np.arange(unemployment.num_rows)/4
      pter_over_time = unemployment.with_columns('Year', year, 'PTER', pter)
      pter_over_time.plot('Year', 'PTER')
      plt.ylim(0,2); # Do not change this line
```



Question 7. Were PTER rates high during the Great Recession (that is to say, were PTER rates particularly high in the years 2008 through 2011)? Assign `highPTER` to `True` if you think PTER rates were high in this period, or `False` if you think they weren't.

[27]: `highPTER = True`

1.2 2. Birth Rates

The following table gives Census-based population estimates for each US state on both July 1, 2015 and July 1, 2016. The last four columns describe the components of the estimated change in population during this time interval. **For all questions below, assume that the word “states” refers to all 52 rows including Puerto Rico and the District of Columbia.**

The data was taken from [here](#). (Note: If the file doesn't download for you when you click the link, you can copy and paste the link address it into your address bar!) If you want to read more about the different column descriptions, click [here](#).

The raw data is a bit messy—run the cell below to clean the table and make it easier to work with.

```
[28]: # Don't change this cell; just run it.
pop = Table.read_table('./DS/nst-est2016-alldata.csv').where('SUMLEV', 40).
    ↪select([1, 4, 12, 13, 27, 34, 62, 69])
pop = pop.relabeled('POPESTIMATE2015', '2015').relabeled('POPESTIMATE2016',
    ↪'2016')
pop = pop.relabeled('BIRTHS2016', 'BIRTHS').relabeled('DEATHS2016', 'DEATHS')
pop = pop.relabeled('NETMIG2016', 'MIGRATION').relabeled('RESIDUAL2016',
    ↪'OTHER')
pop = pop.with_columns("REGION", np.array([int(region) if region != "X" else 0
    ↪for region in pop.column("REGION")]))
pop.set_format([2, 3, 4, 5, 6, 7], NumberFormatter(decimals=0)).show(5)
```

<IPython.core.display.HTML object>

Question 1. Assign `us_birth_rate` to the total US annual birth rate during this time interval. The annual birth rate for a year-long period is the total number of births in that period as a proportion of the total population size at the start of the time period.

Hint: Remember that each row in the `pop` table refers to a state, not the US as a whole.

```
[30]: us_birth_rate = sum(pop.column('BIRTHS')) / sum(pop.column('2015'))
us_birth_rate
```

```
[30]: 0.012358536498646102
```

Question 2. Assign `movers` to the number of states for which the **absolute value** of the **annual rate of migration** was higher than 1%. The annual rate of migration for a year-long period is the net number of migrations (in and out) as a proportion of the population size at the start of the period. The `MIGRATION` column contains estimated annual net migration counts by state.

Hint: `migration_rates` should be a table and `movers` should be a number.

```
[31]: migration_rates = pop.column('MIGRATION') / pop.column('2015')
movers = sum(abs(migration_rates) > 0.01)
movers
```

```
[31]: 9
```

Question 3. Assign `west_births` to the total number of births that occurred in region 4 (the Western US).

Hint: Make sure you double check the type of the values in the `REGION` column and appropriately filter (i.e. the types must match!).

```
[32]: west_births = sum(pop.where('REGION', 4).column('BIRTHS'))
west_births
```

```
[32]: 979657
```

Question 4. In the next question, you will be creating a visualization to understand the relationship between birth and death rates. The annual death rate for a year-long period is the total number of deaths in that period as a proportion of the population size at the start of the time period.

What visualization is most appropriate to see if there is an association between annual birth and death rates across multiple states in the United States?

1. Line Graph
2. Bar Chart
3. Scatter Plot

Assign `visualization` below to the number corresponding to the correct visualization.

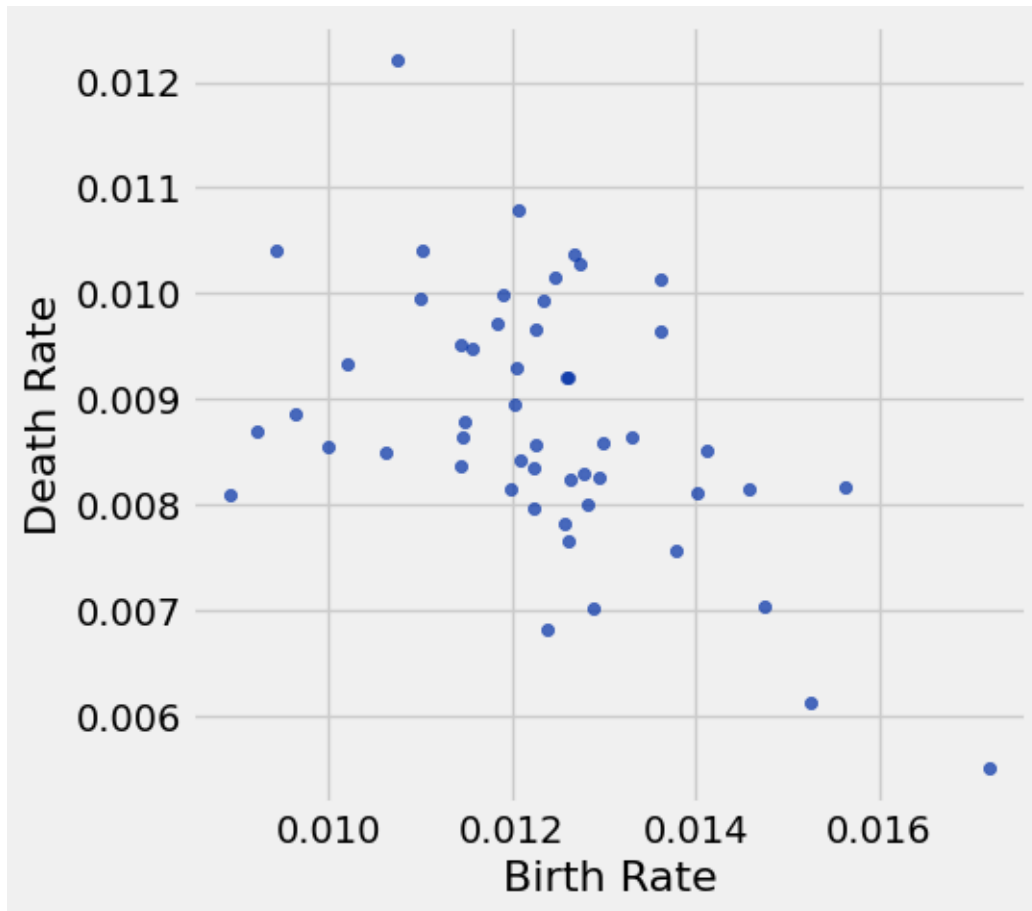
```
[33]: visualization = 3
```

Question 5. In the code cell below, create a visualization that will help us determine if there is an association between birth rate and death rate during this time interval. It may be helpful to create an intermediate table containing the birth and death rates for each state.

Things to consider:

- What type of chart will help us illustrate an association between 2 variables?
- How can you manipulate a certain table to help generate your chart?

```
[34]: birth_rates_2015 = pop.column('BIRTHS') / pop.column('2015')
death_rates_2015 = pop.column('DEATHS') / pop.column('2015')
pop_with_rates = pop.with_columns(
    'Birth Rate', birth_rates_2015,
    'Death Rate', death_rates_2015
)
pop_with_rates.scatter('Birth Rate', 'Death Rate')
```



Question 6. True or False: There is an association between birth rate and death rate during this time interval.

Assign `assoc` to `True` or `False` in the cell below.

```
[35]: assoc = True
```

1.3 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit.

Please save before exporting!

```
[ ]: # should change the directory and file name matching to yours
!jupyter nbconvert './DS/lab05(SSU).ipynb' --to pdf
```